11b. Introduction to Multithreading

Martin Alfaro
PhD in Economics

INTRODUCTION

A correct implementation of multithreading requires a basic understanding of how computers execute instructions. In particular, it's essential to understand the procedure in light of data dependencies, where one operation relies on the output of a previous one. If these dependencies aren't properly managed, multithreading can lead to incorrect results and introduce several forms of unsafe code.

This section introduces the preliminary concepts needed to reason about these issues, laying the foundation for the more practical discussions that follow. **The emphasis here is on explanation rather than implementation**. Accordingly, many of the macros and functions presented won't be reused elsewhere in this book.

> ⚠️ **Warning!** - Loaded Package
> All the scripts below assume that ou've executed the line `using Base.Threads`. This allows direct access to the macro `@spawn`.

NATURE OF COMPUTATIONS

Operations can be broadly classified based on their data dependency. A **dependent operation** is one whose outcome is influenced by the result of another operation. In such cases, the order of execution is critical, because changing the sequence can alter the final outcome. In contrast, an **independent operation** always produces the same result, regardless of the order in which it's executed relative to other operations.

The following code gives rise to a dependent or independent operation, depending on which values `job_B` sums.

DEPENDENT OPERATIONS

```
job_A()  = 1 + 1
job_B(A) = 2 + A

function foo()
    A = job_A()
    B = job_B(A)

    return A,B
end
```

INDEPENDENT OPERATIONS

```
job_A()  = 1 + 1
job_B()  = 2 + 2

function foo()
    A = job_A()
    B = job_B()

    return A,B
end
```

Furthermore, regardless of their dependency status, operations can be computed either sequentially or concurrently. A **sequential** procedure involves executing operations one after the other, ensuring each completes before the next one begins. Conversely, **concurrency** allows multiple operations to be processed simultaneously, opening up opportunities for parallel execution.

Like most programming languages, **Julia defaults to a sequential execution**. This is a deliberate design choice that prioritizes result correctness, as **concurrent execution of dependent operations can yield incorrect results if mishandled**. The issue arises because concurrency introduces non-determinism in the execution order, which can lead to timing inconsistencies when reading and writing data. A sequential approach precludes this possibility by guaranteeing a predictable order of execution.

Despite its advantages regarding safety, a sequential approach can be quite inefficient for independent tasks: by restricting computations to one at a time, computational resources may go underutilized. In contrast, **a simultaneous approach allows for operations to be calculated in parallel, thereby fully utilizing all the available computational resources**. This can lead to significant reductions in computation time.

Because most programming languages default to sequential execution, certain nuances of concurrent programming can be difficult to grasp (e.g., concurrency doesn't necessarily imply simultaneity). Such misunderstandings can lead to flawed program design. To address this, we next revisit the topic in light of the fundamental concepts of tasks and threads.

TASKS AND THREADS

When computing an operation, Julia internally defines a set of instructions to be processed. This is achieved through the concept of **tasks**. To be computed, a task must be assigned to **a computer thread**. Since a single task runs on exactly one thread at a time, **the number of threads available on your computer determines how many tasks can be computed simultaneously**. Importantly, every Julia session starts with a predefined pool of available threads. By default, **a fresh installation of Julia configures each session to run on a single thread**, regardless of your machine's hardware capabilities.

Imagine two workers, A and B, each responsible for a different task. To make the analogy concrete, think of them as employees in a company.

To explain concurrency and distinguish it from simultaneity, it helps to start with a strictly single-threaded setting. Consider two workers A and B, each responsible for a different task. To establish a parallelism that helps build intuition, think of them as employees working for a company. B's job consists of performing the same operation continuously for a certain period of time. In the code, this is represented by summing `1+1` repeatedly for one second. For its part, A's job consists of waiting for some delivery, which will arrive after a certain period of time. In the code, this job is represented by performing no computations for two seconds, simulated by calling the function `sleep(2)`.

The following code snippet defines functions to capture each worker's job.

A'S JOB

```julia
function job_A(time_working)
    sleep(time_working)        # do nothing (waiting for some delivery in the example)

    println("A completed his task")
end
```

B'S JOB

```julia
function job_B(time_working)
    start_time = time()

    while time() - start_time < time_working
        1 + 1                  # compute `1+1` repeatedly during `time_working` seconds
    end

    println("B completed his task")
end
```

Due to the lazy nature of function definitions, these code snippets merely create a blueprint for a set of operations. No computation is performed. It's only when we call these functions by adding lines like `job_A(2)` and `job_B(1)` that the operations are sent for execution.

To make the underlying execution model transparent, let's adopt a lower-level approach. In particular, we define the function calls `job_A(2)` and `job_B(1)` as tasks. As shown below, tasks aren't merely abstractions to organize our discussion, but actual constructs in Julia's codebase.

DEFINING TASKS

```
A = @task job_A(2)      # A's task takes 2 seconds
B = @task job_B(1)      # B's task takes 1 second
```

Once a task is defined, the first step toward execution is **scheduling**. This places the task into the queue of operations, waiting to be executed by the computer's processor. As soon as a thread becomes available, the computer begins its computation.

Importantly, multiple tasks can be *processed* concurrently, without implying that they'll be *computed* simultaneously. Indeed, this is the case in a single-thread session. The distinction can be understood through an analogy with juggling: a juggler manages multiple balls at the same time, but only holds one ball at any given moment. Similarly, multiple tasks can be processed simultaneously, even when only one is being executed on the CPU.

The implication of this statement is that true parallelism isn't feasible in single-threaded sessions. Nonetheless, concurrency can still improve efficiency through **task switching**, which is enabled by a **task-yielding** mechanism: when a task becomes idle (e.g., when waiting for input or data), it can voluntarily relinquish control of the thread, allowing other tasks to utilize the thread's time. By fostering a cooperative approach, concurrency ensures plenty of computer resource utilization at any given time.

In the following, we describe this mechanism in more detail.

SEQUENTIAL AND CONCURRENT COMPUTATIONS

While code in Julia executes sequentially by default, **tasks are designed to run concurrently**. This means that if we want to enforce a sequential execution of tasks, we must instruct Julia to run them one at a time. The way to do this is by introducing a "wait" instruction immediately after scheduling a task. By waiting, we ensure that the task finishes before proceeding with the rest of the program.

The code snippet below demonstrates this mechanism by introducing the functions `schedule` and `wait`.

JULIA'S DEFAULT PROCEDURE (SEQUENTIAL)

```
A = job_A(2)            # A's task takes 2 seconds
B = job_B(1)            # B's task takes 1 second
```

SEQUENTIAL

```
A = @task job_A(2)      # A's task takes 2 seconds
B = @task job_B(1)      # B's task takes 1 second

schedule(A) |> wait
schedule(B) |> wait
```

CONCURRENT

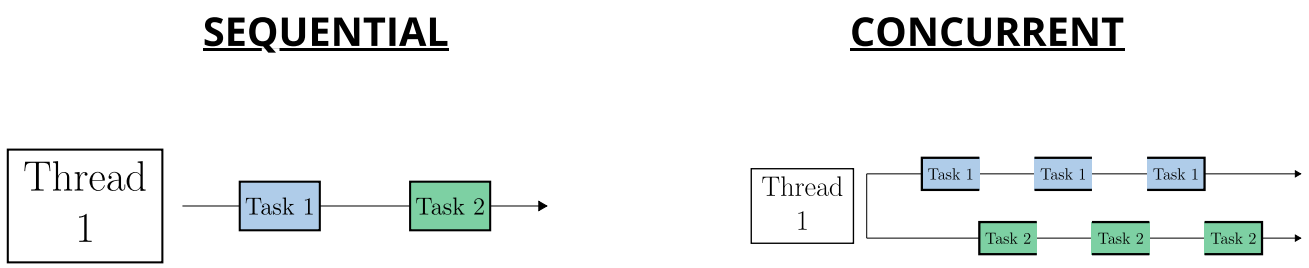
```
A = @task job_A(2)      # A's task takes 2 seconds
B = @task job_B(1)      # B's task takes 1 second

(schedule(A), schedule(B)) .|> wait
```

Note that `wait` was added even in the concurrent case and after both tasks were scheduled. The purpose is to pause the main program flow, until each scheduled task has completed. In this way, we prevent subsequent code from executing prematurely.

The example reveals the benefits of task switching under concurrency: although only one task can run at any moment, task A can yield control of the thread to task B when it becomes idle. In the code, the idle state is simulated by the function `sleep`, during which the computer performs no operations. Once task A becomes idle, its state is saved, allowing it to eventually resume execution from where it left off. In the meantime, task B can use that thread's processing time, explaining why task B finishes first.

By taking turns efficiently and sharing the single available thread, tasks make the most of the CPU's processing power. This contrasts with a sequential approach, where task A must finish before moving to the next task. The difference is reflected in the execution times, resulting in 2 seconds for the concurrent approach and 3 seconds for the sequential one.



Examples of idle states emerge naturally in real programs. A common case is when a program waits for user input, such as a keystroke or mouse click. It can also arise when browsing the internet, where the CPU may idle while waiting for a server to send data. Task switching is so ubiquitous in certain

contexts that we often take it for granted. For instance, I bet you've never wondered whether you could use the computer while a document prints in the background!

Notice, though, that concurrency with a single thread offers no benefits if both tasks require active computations. In that scenario, the CPU would be fully utilized, leaving no chance for task switching. As a result, the sequential and concurrent approaches would become equivalent. In our example, this possibility occurs when both task A and task B compute `1+1` repeatedly, leading to an identical execution time of 3 seconds in either approach.

DEFINING TASK - SAME FOR A AND B

```
function job(name_worker, time_working)
    start_time = time()

    while time() - start_time < time_working
        1 + 1                   # compute `1+1` repeatedly during `time_working` seconds
    end

    println("$name_worker completed his task")
end
```

SEQUENTIAL

```
function schedule_of_tasks()
    A = @task job("A", 2)       # A's task takes 2 seconds
    B = @task job("B", 1)       # B's task takes 1 second

    schedule(A) |> wait
    schedule(B) |> wait
end
```

CONCURRENT

```
function schedule_of_tasks()
    A = @task job("A", 2)      # A's task takes 2 seconds
    B = @task job("B", 1)      # B's task takes 1 second

    (schedule(A), schedule(B)) .|> wait
end
```

MULTITHREADING

The procedure just outlined showed that **when a task is scheduled, the computer attempts to find an available thread to run it**. Our discussion of concurrency so far has assumed that the session was running on a single thread. However, the situation changes **when starting a session with multiple threads, in which case true parallel code execution becomes possible**. This setting is simply referred to as **multithreading**.

To build intuition, let's revisit the case where both workers A and B perform meaningful computations. The only change introduced is that Julia's session now starts with more than one available thread. For the concurrent version, we also specify that the tasks are "non-sticky". This establishes that a task can run on any available thread, rather than the one it started on. Such flexibility allows for more efficient resource allocation.

Once there's more than one thread available, concurrency implies simultaneity: each task can run on a different thread. This is why task B finishes first in the following implementation.

SEQUENTIAL

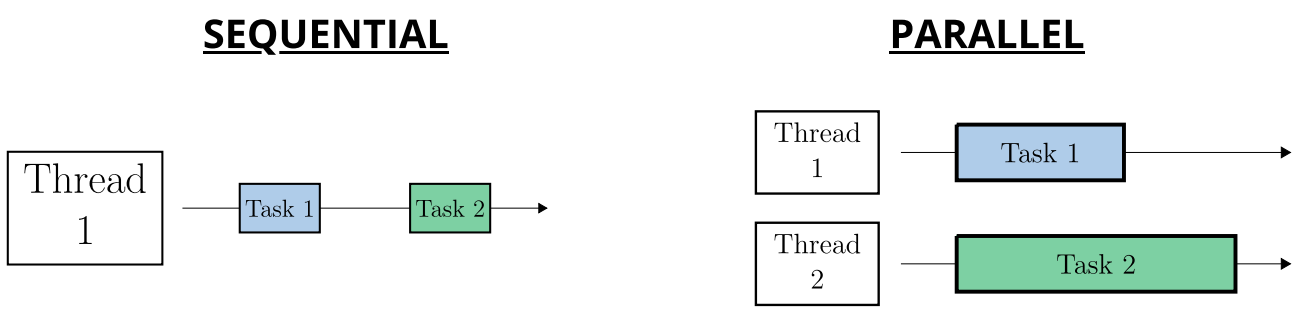
```julia
function schedule_of_tasks()
    A = @task job("A", 2)                        # A's task takes 2 seconds
    B = @task job("B", 1)                        # B's task takes 1 second

    schedule(A) |> wait
    schedule(B) |> wait
end
```

PARALLEL

```julia
function schedule_of_tasks()
    A = @task job("A", 2) ; A.sticky = false      # A's task takes 2 seconds
    B = @task job("B", 1) ; B.sticky = false      # B's task takes 1 second

    (schedule(A), schedule(B)) .|> wait
end
```



To see the benefits of multithreading in action, let's compare Julia's default implementation (sequential) with a multithreaded one. The macro `@spawn`, which will be covered in the next section, offers a simple way to run tasks in a multithreaded environment. Essentially, it's equivalent to creating and scheduling a non-sticky task.

SEQUENTIAL (JULIA'S DEFAULT PROCEDURE)

```julia
function schedule_of_tasks()
    A = job("A", 2)             # A's task takes 2 seconds
    B = job("B", 1)             # B's task takes 1 second
end
```

MULTITHREADING

```julia
function schedule_of_tasks()
    A = @spawn job("A", 2)      # A's task takes 2 seconds
    B = @spawn job("B", 1)      # B's task takes 1 second

    (A,B) .|> wait
end
```

THE IMPORTANCE OF WAITING FOR THE RESULTS

Before concluding this section, it's worth stressing a crucial point: you must always instruct your program to wait for operations to complete before proceeding. This holds true even for concurrent computations. **Failing to wait may produce incorrect results, even in a single-threaded**

environment.

To illustrate this, consider mutating a vector in a single-threaded session, with a one-second delay for each value update. If we don't wait for the mutation to finish, any subsequent operation will be based on the vector's value at the moment it's accessed. This value doesn't necessarily reflect its final mutated state, but merely its value at the moment of reference.

For instance, suppose mutate the vector `x = [0,0,0]` into `x = [1,1,1]` and then sum all its value. Let's begin considering Julia's default sequential execution. This ensures that the mutation completes before continuing with the sum.

MUTATION - JULIA'S DEFAULT APPROACH

```julia
# Description of job
function job!(x)
    for i in 1:3
        sleep(1)      # do nothing for 1 second
        x[i] = 1      # mutate x[i]

        println("`x` at this moment is $x")
    end
end

# Execution of job
function foo()
    x = [0, 0, 0]

    job!(x)           # slowly mutate `x`

    return sum(x)
end

output = foo()
println("the value stored in `output` is $(output)")
```

Let's now consider the same implementation but through tasks. In particular, we define a task that performs the mutation as follows.

DEFINE TASK

```
function job!(x)
    @task begin
        for i in 1:3
            sleep(1)    # do nothing for 1 second
            x[i] = 1    # mutate x[i]

            println("`x` at this moment is $x")
        end
    end
end
```

The following snippets show the consequences of waiting versus not waiting for the mutation task to complete.

NOT WAITING

```
function foo()
    x = [0, 0, 0]

    job!(x) |> schedule             # define job, start execution, don't wait for job to be
done

    return sum(x)
end

output = foo()
println("the value stored in `output` is $(output)")
```

WAITING

```
function foo()
    x = [0, 0, 0]

     job!(x) |> schedule |> wait       # define job, start execution, only continue when
finished

    return sum(x)
end

output = foo()
println("the value stored in `output` is $(output)")
```

Without a `wait` call, the main program schedules the mutation task and immediately proceeds to the next line. Since the task contains a `sleep` instruction, the mutation hasn't yet begun when `x` is accessed for being summed. This results in the use of `x`'s original value `[0,0,0]`, determining that `sum(x)` provides `0` as output.